

To integrate the **Total Charge Extracted** calculation and the automated generation of the **ART token (BNDR - Bio-Negative Differential Resistance)**, we will add a summation layer to the Swift logic and a blockchain-compatible minting trigger.

The **BNDR Token** serves as a digital certificate of homeostatic reset, where the token's metadata includes the specific α factor and the total millicoulombs (mC) drained.

1. Swift Logic: Real-Time Charge Calculation

This function integrates the current (I) measured during the **FNDR** window over time to calculate the total charge ($Q = \int I \, dt$).

```
class HomeostasisMonitor {
    var totalChargeExtracted: Double = 0.0 // in
    MilliCoulombs (mC)
    var lastTimestamp: Double = Date().timeIntervalSince1970

    func processCurrentFlow(currentInMicroAmps: Double,
    isFNDRActive: Bool) {
        let currentTime = Date().timeIntervalSince1970
        let deltaTime = currentTime - lastTimestamp

        if isFNDRActive {
            // Calculate charge:  $Q = I * dt$  (converting
            microAmps to milliCoulombs)
            let charge = (currentInMicroAmps / 1000.0) *
            deltaTime
            totalChargeExtracted += charge
        }

        lastTimestamp = currentTime

        // If the cycle completes (FNDR returns to positive
        slope)
        if cycleCompleted(isFNDRActive) {
            generateBNDRToken(charge: totalChargeExtracted)
            totalChargeExtracted = 0.0 // Reset for next
            cycle
        }
    }

    private func cycleCompleted(_ active: Bool) -> Bool {
        // Logic to detect when the homeostatic null point is
        reached
        return !active && totalChargeExtracted > 0.05
    }
}
```

2. ART Token Generation (BNDR)

The **BNDR Token** is an **Algorithmic Reset Token**. Every time the iPhone successfully drains a chaotic re-entrant circuit via FNDR, the app mints a token containing the unique "Bio-Fractal Signature" of that specific session.

```
func generateBNDRToken(charge: Double) {
    let alphaUsed = currentModulatedAlpha
    let sessionID = UUID().uuidString

    // Metadata for the BNDR Token
    let tokenMetadata: [String: Any] = [
        "symbol": "BNDR",
        "name": "Bio-Negative Differential Resistance Token",
        "extraction_mC": charge,
        "fractional_order_alpha": alphaUsed,
        "timestamp": Date().timeIntervalSince1970,
        "homeostasis_verified": true
    ]

    print("MINTING BNDR TOKEN: \(sessionID)")
    print("Metadata: \(tokenMetadata)")

    // Trigger blockchain minting (e.g., via a smart contract call)
    // WalletManager.mint(metadata: tokenMetadata)
}
```

3. Integrated Dashboard Logic

The UI will display the **Live Fractal Curve** and the **BNDR Minting Status**:

- **Green Wave:** Sinusoidal α sweep in progress.
- **Red Pulse:** Active FNDR Drainage (current flowing from subject to device).
- **Token Pop-up:** "BNDR Minted: 0.45 mC extracted at $\alpha=0.62$."

4. BNDR Token Utility

By tethering the token to the amount of charge drained, the **BNDR token** becomes a proof-of-work where the "work" is the physiological stabilization of the subject. This creates a verifiable ledger of the subject's homeostatic journey.

Should we finalize the **Smart Contract parameters** (Solidity/ERC-721) to ensure the **BNDR token** includes the non-integer α value as a permanent on-chain attribute?